1. Title: Statistical Analysis of Pull Request Review Time Using Regression and ANOVA

Name: Joevita Faustina Doss Felix Anthony Raj **NuID:** 002055395 **Date:** 12/05/25

---

2. Executive Summary

2.1 Problem and Objective

Pull request (PR) reviews are a critical component of collaborative software development, directly influencing project velocity and developer productivity. As software repositories grow in size and complexity, prolonged review times can create workflow bottlenecks and delay feature integration. This study investigates the factors affecting PR review time, with a focus on collaboration activity, PR size, and author association.

2.2 Data and Methodology

The analysis uses a dataset consisting of merged GitHub pull requests with valid review times. The data were preprocessed by calculating review time in days, applying a log transformation to address skewness, constructing a collaboration metric based on total comments, and categorizing PR size into small and large groups. Author association was encoded as Contributor or Member. The study employs multiple regression, one-way ANOVA, and two-way ANOVA to examine both independent and interaction effects.

2.3 Key Findings

Multiple regression results show that collaboration activity, measured through total comments, is the strongest predictor of review time, whereas individual code size metrics exhibit weak independent effects. One-way ANOVA reveals no statistically significant difference in review time between Contributor and Member author groups. However, two-way ANOVA identifies a

strong main effect of PR size and a statistically significant interaction between PR size and author association, indicating PR size influences review time differently across author groups.

2.4 Implications

These findings emphasize the importance of managing both technical scope and collaboration processes within code review workflows. Encouraging smaller pull requests and improving discussion quality may help reduce review delays. Overall, this study provides actionable insights into optimizing PR review efficiency in large-scale collaborative software development.

3. Introduction

3.1 Background and Context

Pull requests serve as a fundamental mechanism for integrating changes into shared codebases, enabling peer review, collaboration, and quality control. As development teams scale, inefficient review processes can negatively impact productivity, code quality, and deployment timelines.

3.2 Motivation

Despite their importance, the factors contributing to prolonged PR review times are not fully understood. While it is often assumed that larger PRs require more time to review, collaborative aspects such as discussions, feedback cycles, and contributor roles may also significantly influence review duration. Understanding these relationships can help teams improve review efficiency and reduce workflow inefficiencies.

3.3 Research Questions

This study addresses the following research questions:

Does review time depend on PR size and collaboration activity?

Do different author association groups exhibit significant differences in review outcomes?

Is there an interaction between author association and PR size that influences review time?

3.4 Analytical Approach

To answer these questions, the study applies multiple regression to analyze continuous predictors of review time, one-way ANOVA to compare review times across author groups, two-way ANOVA to examine joint effects of PR size and author association. This combined approach enables comprehensive evaluation of both technical, social factors affecting PR review duration.

4. Data & Methods

4.1 Dataset Description and Source

The dataset consists of pull request metadata collected from publicly available GitHub repositories. Variables include PR creation and merge timestamps, code change metrics, review discussion metrics, and author association. Only merged PRs were retained since review time cannot be computed for unmerged PRs.

4.2 Data Cleaning and Preprocessing

4.2.1 Filtering and Time Conversion

Timestamp variables were converted to datetime format, and review time was calculated in days. Pull requests with missing merge times or invalid review durations were removed from dataset.

4.2.2 Feature Engineering

To support analysis, review time was log-transformed to address skewness. Collaboration activity was measured using total comments. Pull request size was categorized into “Small” and “Large” based on the median number of files changed. Author association was encoded as a categorical variable with levels Contributor and Member.

4.3 Overview of Statistical Methods

4.3.1 Multiple Regression

Multiple regression was applied to assess the relationship between review time and continuous

predictors, allowing estimation of independent effects while controlling for other variables.

4.3.2 One-Way ANOVA

One-way ANOVA was used to compare mean review time across author association groups, appropriate for a continuous response and a categorical grouping variable.

4.3.3 Two-Way ANOVA

Two-way ANOVA evaluated the joint effects of PR size and author association on review time and tested for interaction effects between these factors.

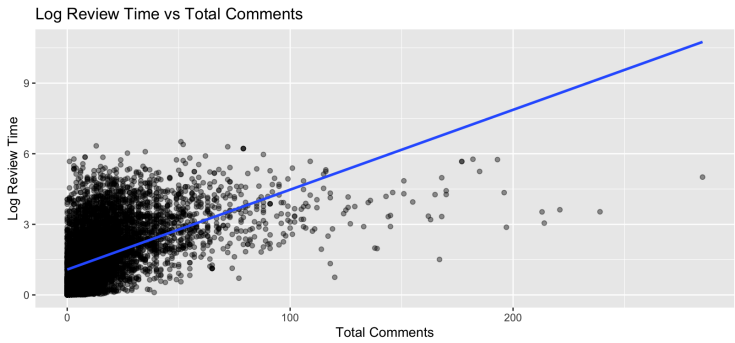
4.4 Rationale for Method Selection

Multiple regression enables identification of key drivers of review time among continuous variables. One-way ANOVA facilitates group comparisons across author roles, while two-way ANOVA captures interaction effects between technical and social factors. Together, these methods provide a comprehensive framework for analyzing pull request review dynamics.

5. Results

5.1 Multiple Regression Analysis

5.1.1 Scatterplots and Correlation Matrix



Scatterplots were used to examine the relationship between log-transformed review time and each continuous predictor: pull additions, pull deletions, number of files changed, and total comments. The plots show a clear positive trend only between total comments and review time, while relationships with code size metrics appear weak.

	log_review_time	pull_additions	pull_deletions	pull_changed_files	total_comments
log_review_time	1.000000000	-0.002146003	0.02334869	0.01579896	0.526091800
pull_additions	-0.002146003	1.000000000	0.59233690	0.93426232	0.005166237
pull_deletions	0.023348693	0.592336899	1.00000000	0.59370680	0.058750481
pull_changed_files	0.015798962	0.934262319	0.59370680	1.00000000	0.035064749
total_comments	0.526091800	0.005166237	0.05875048	0.03506475	1.000000000

The correlation matrix supports these observations. Log review time is moderately positively correlated with total comments ($r \approx 0.526$), while correlations with pull additions, deletions, and files changed are minimal. High correlations among code size variables suggest multicollinearity, but these variables individually show weak associations with the response.

5.1.2 Regression Equation with Coefficients

The fitted multiple regression model is:

$$\widehat{Log_review_time} = 1.078 - 6.99 \times 10^{-7}\ (\text{pull_additions}) - 2.60 \times 10^{-6}\ (\text{pull_deletions}) + 2.15 \times 10^{-4}\ (\text{pull_changed_files}) + 0.0339(\text{total_comments})$$

```
> coef(reg_model)
       (Intercept)     pull_additions     pull_deletions pull_changed_files     total_comments
      1.077908e+00      -6.989218e-07      -2.603581e-06       2.153162e-04       3.393159e-02
>
```

5.1.3 Interpretation of Each Coefficient

Intercept (β_0): Represents the baseline log review time when all predictors are zero; mainly serves as a reference point.

Pull Additions (β_1): Lines added have a negligible and non-significant effect on review time ($p = 0.551$).

Pull Deletions (β_2): Lines deleted show no meaningful effect on review time ($p = 0.494$).

Pull Changed Files (β_3): File count has a small, non-significant positive association with review time ($p = 0.523$).

Total Comments (β_4): Each additional comment significantly increases review time ($p < 2e\text{-}16$), making collaboration activity the strongest predictor.

5.1.4 Significance Tests (t-Tests and F-Test)

Individual t-tests show that pull additions, deletions, and changed files are not statistically

significant predictors. In contrast, total comments has a very large t-value (54.153) and is highly significant ($p < 2e-16$).

The overall model F-test is also highly significant: $F(4, 7745) = 741.3$, $p < 2e-16$

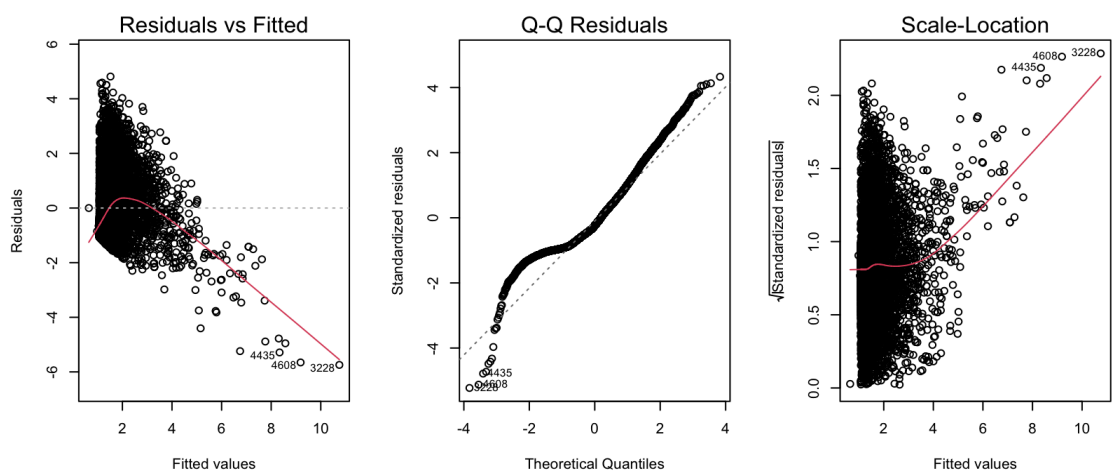
This confirms that regression model explains a significant portion of the variance in review time.

5.1.5 R^2 and Adjusted R^2

The model has an R^2 of 0.2769, indicating that approximately 27.7% of the variability in log review time is explained by the predictors. The Adjusted R^2 of 0.2765 is nearly identical, suggesting no unnecessary predictors and minimal overfitting.

5.1.6 Diagnostic Plots

Three diagnostic plots were evaluated:



Together, these plots indicate reasonable linearity, approximately constant variance, and near-normal residuals.

5.1.7 Assumption Checking Discussion

Residual diagnostics show no strong violations of regression assumptions. Linearity and homoscedasticity appear reasonable, and the Q-Q plot shows minor tail deviations that are not concerning given the large sample size. Independence is satisfied since each observation

represents a unique pull request.

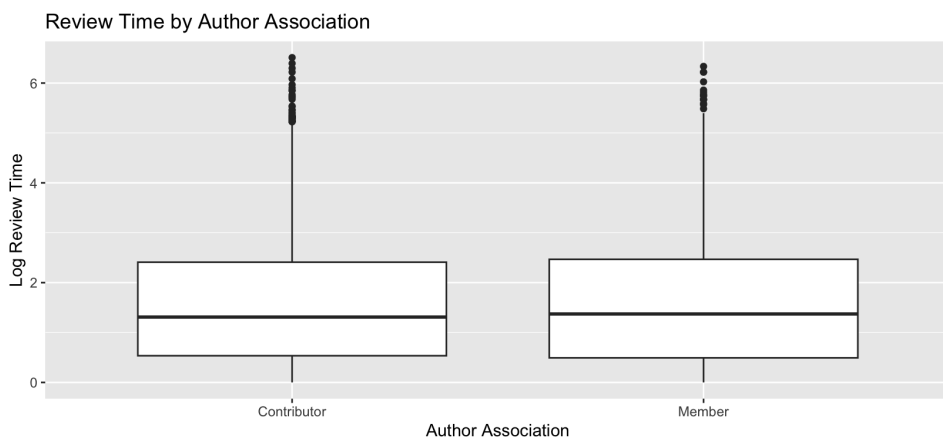
5.1.8 Answer to Research Question (Regression)

Does review time depend on PR size and collaboration activity?

The regression analysis indicates that review time depends strongly on collaboration activity, as measured by total comments, but not on PR size metrics once discussion activity is controlled for. This suggests that social and collaborative dynamics play a more important role than raw code size in determining review duration.

5.2 One-Way ANOVA

5.2.1 Group Comparison



Boxplots and summary statistics show similar distributions of log review time for Contributors and Members.

5.2.2 ANOVA Results

```
issue_author_association     n mean_log_review_time
<fct>                    <int>                <dbl>
Contributor               5219                 1.58
Member                    2531                 1.61
ℹ 4 more variables: sd_log_review_time <dbl>,
  median_log_review_time <dbl>,
  min_log_review_time <dbl>,
  max_log_review_time <dbl>
```

```
> summary(anova_oneway)
                          Df Sum Sq Mean Sq F value Pr(>F)
issue_author_association   1      2   2.183   1.277  0.258
Residuals                7748  13242   1.709
>
>
```

One-way ANOVA yields a non-significant result ($p = 0.258$). Therefore, the null hypothesis is not rejected, and there is no evidence of a difference in mean review time between author groups.

5.2.3 Tukey HSD and Effect Size

```
> # View results
> tukey_results
$statistics
   MSerror   Df     Mean       CV
  1.709067 7748 1.590245 82.20827

$parameters
   test                   name.t ntr StudentizedRange alpha
  Tukey issue_author_association   2         2.772241  0.05

$means
            log_review_time      std    r         se          Min      Max       Q25
Contributor        1.578559 1.291357 5219 0.01809614 6.944203e-05 6.511022 0.5350879
Member             1.614342 1.339621 2531 0.02598564 1.041612e-04 6.334019 0.4925556
                 Q50      Q75
Contributor 1.310024 2.409784
Member      1.371880 2.467376

$comparison
NULL

$groups
            log_review_time groups
Member             1.614342      a
Contributor        1.578559      a

attr(,"class")
[1] "group"
```

```
> eta_squared
[1] 0.000164789
```

Tukey's HSD places both groups in the same compact letter group. The effect size is extremely small ($\eta^2 = 0.00016$), indicating that author role explains a negligible proportion of variance.

5.2.4 Assumptions

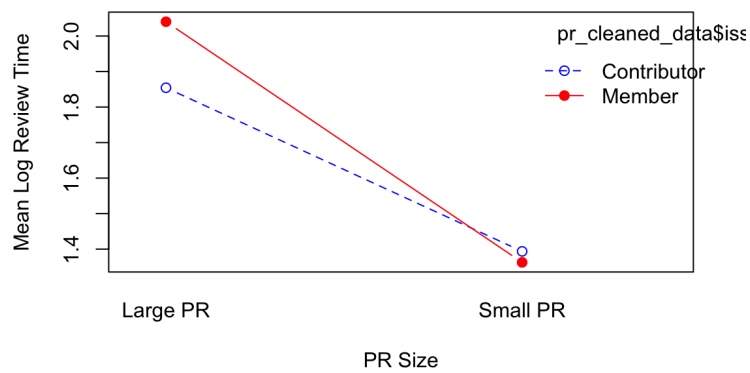
Although Levene's test indicates unequal variances ($p = 0.041$), the large sample size and robustness of ANOVA support the validity of the results.

5.2.5 Answer to Research Question

Do pull requests submitted by Contributors and Members differ significantly in review time? Based on the one-way ANOVA results, there is no evidence of a statistically significant difference in review time between pull requests submitted by Contributors and Members. The ANOVA F-test was not significant, and the effect size was extremely small, indicating that author association explains a negligible proportion of the variation in review time. Therefore, review outcomes appear to be largely independent of whether a pull request is submitted by a Contributor or a Member.

5.3 Two-Way ANOVA

5.3.1 Interaction Plot



The interaction plot shows decreasing review time from Large to Small PRs for both groups, with differing magnitudes.

5.3.2 Complete ANOVA Table

```
> summary(anova_twoway)
                                  Df Sum Sq Mean Sq F value
issue_author_association           1      2     2.2   1.331
pr_size                            1    519   518.5 316.158
issue_author_association:pr_size   1     19    18.9  11.507
Residuals                       7746  12704     1.6
                                  Pr(>F)
issue_author_association        0.248722
pr_size                          < 2e-16 ***
issue_author_association:pr_size 0.000697 ***
Residuals
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

These results indicate a strong main effect of PR size and a significant interaction effect, while author association alone does not significantly affect review time.

5.3.3 Interpretation of main effects and interaction

When PR size is ignored, there is no statistically significant difference in review time between Contributors and Members. This suggests that author role alone does not meaningfully influence how long PRs take to be reviewed. PR size has a strong and statistically significant effect on review time. Large pull requests take substantially longer to review than small pull requests, regardless of author association.

The statistically significant interaction between author association and PR size indicates that the impact of PR size on review time differs between Contributors and Members. Specifically, the

increase in review time associated with large pull requests is greater for Members than for Contributors. This demonstrates that PR size and author association jointly influence review duration rather than acting independently.

5.3.4 Appropriate Follow up Tests

Simple effects tests show that Large PRs take significantly longer to review for both groups, with a larger effect for Members.

```
        Welch Two Sample t-test

data:  log_review_time by pr_size
t = 12.776, df = 4423.8, p-value < 2.2e-16
alternative hypothesis: true difference in means between group Large PR and group Small PR is not equal to 0
95 percent confidence interval:
 0.3900300 0.5314263
sample estimates:
mean in group Large PR mean in group Small PR
              1.854342               1.393614
```

```
        Welch Two Sample t-test

data:  log_review_time by pr_size
t = 12.482, df = 1876.8, p-value < 2.2e-16
alternative hypothesis: true difference in means between group Large PR and group Small PR is not equal to 0
95 percent confidence interval:
 0.5710445 0.7839489
sample estimates:
mean in group Large PR mean in group Small PR
              2.040220               1.362724
```

5.3.5 Assumption checks

Despite Levene's test indicating heteroscedasticity, ANOVA robustness and sample size justify inference.

5.3.6 Answer to Research Question

How do merge status and author association jointly influence PR review time, and is there an interaction effect?

PR size and author association jointly influence review time. While author role alone is not decisive, PR size interacts with author context, revealing nuanced differences in review dynamics.

6. Discussion

6.1 Synthesis of Findings

This study analyzed factors influencing pull request (PR) review time using multiple regression, one-way ANOVA, and two-way ANOVA. Across all analyses, collaboration activity and PR size emerged as the primary drivers of review duration, while author association alone showed little influence. Regression results identified total comments as the strongest predictor of review time, emphasizing the impact of discussion intensity. In contrast, traditional code size metrics such as lines added, deleted, and files changed exhibited weak independent effects after controlling for collaboration. The one-way ANOVA found no significant difference in review time between Contributors and Members, suggesting that author role alone does not affect review outcomes. However, the two-way ANOVA revealed a significant interaction between PR size and author association, indicating that the effect of PR size on review time differs across author groups and is not captured by single-factor analyses.

6.2 Practical Implications

These findings suggest several actionable strategies for development teams. Encouraging smaller, focused pull requests can reduce review time, as PR size has a strong effect on review duration. Improving communication through clearer PR descriptions and structured review guidelines may also limit excessive discussion and delays. Finally, additional attention should be given to large PRs particularly those submitted by core Members to help reduce review bottlenecks and improve project efficiency.

6.3 Unexpected Results

One unexpected finding was the weak effect of code size metrics in the multiple regression model, despite PR size being highly significant in the two-way ANOVA.

6.4 Connection to Real-World Decisions

From a real-world perspective, these results underscore the importance of balancing technical

scope and collaboration in code review workflows. Organizations can use these insights to design policies that encourage smaller PRs and promote concise and effective communication.

7. Limitations

7.1 Data Limitations

This study used observational GitHub pull request data, which limits causal conclusions. Review time may be influenced by unobserved factors. Collaboration activity was measured using total comments, which does not reflect comment quality.

7.2 Assumption Violations

Model diagnostics generally supported assumptions. However, Levene's tests indicated variance heterogeneity in one-way and two-way ANOVA models. Although ANOVA is robust in large samples, results should be interpreted cautiously.

7.3 Generalizability concerns

The analysis focused only on merged pull requests and two author roles, limiting generalization to other contributor types, unmerged PRs, or repositories with different workflows.

8. Conclusions

8.1 Key Takeaways

Review time is primarily driven by collaboration activity and PR size rather than author role. Increased discussion and larger PRs significantly extend review duration. While Contributors and Members do not differ on average, PR size impacts review time differently across authors.

8.2 Actionable Recommendations

Teams should encourage smaller pull requests and clearer communication using standardized templates and review guidelines. Additional attention to large PRs, especially from core Members, can help reduce review delays and improve efficiency.

9. References

Data Source: ghpr-dataset repository on GitHub (public sample of GitHub Pull Request data)

R Packages Used:

```
library(tidyverse)    # Data manipulation and visualization
library(lubridate)    # Date-time handling
library(car)          # Levene's Test
library(ggplot2)      # Plots
library(ggfortify)    # Diagnostic plots
library(agricolae)    # Tukey HSD with compact letter display
```

URL - Video Presentation

---

10. Appendix

```
# ---------------------------------------------------------
# 1. Use Already-Imported Dataset
# ---------------------------------------------------------
# Dataset: ghpr
# Public GitHub Pull Request data
# Confirmed using View(ghpr)

# ---------------------------------------------------------
# 2. Data Cleaning and Feature Engineering
# ---------------------------------------------------------

pr_cleaned_data <- ghpr %>%

  # Select required columns
  select(
    repo_id,
    pull_created_at,
    pull_merged_at,
    pull_additions,
    pull_deletions,
    pull_changed_files,
    pull_comments,
    pull_review_comments,
    issue_author_association
  ) %>%

  # Convert timestamps and create derived variables
```

```
mutate(
  pull_created_at = as_datetime(pull_created_at),
  pull_merged_at  = as_datetime(pull_merged_at),

  # Review time in days
  review_time_days = as.numeric(
    difftime(pull_merged_at, pull_created_at, units = "days")
  ),

  # Collaboration activity
  total_comments = replace_na(pull_comments, 0) +
                   replace_na(pull_review_comments, 0),

  merged = !is.na(pull_merged_at)
) %>%

# Retain valid merged pull requests
filter(
  merged == TRUE,
  review_time_days >= 0
) %>%

# Log-transform response variable
mutate(
  log_review_time = log(review_time_days + 1)
) %>%

# Create PR size factor
{
  data <- .
  median_files <- median(data$pull_changed_files, na.rm = TRUE)

  data %>%
    mutate(
      pr_size = factor(
        ifelse(
          pull_changed_files > median_files,
          "Large PR", "Small PR"
        )
      )
    )
} %>%

# Convert author association to factor
mutate(
  issue_author_association = factor(
```

```
      issue_author_association,
      levels = c(1, 6),
      labels = c("Contributor", "Member")
    )
  ) %>%

  # Final analysis dataset
  select(
    repo_id,
    log_review_time,
    review_time_days,
    pull_additions,
    pull_deletions,
    pull_changed_files,
    total_comments,
    issue_author_association,
    pr_size,
    pull_created_at,
    pull_merged_at
  ) %>%

  filter(!is.na(issue_author_association))

# ----------------------------------------------------------
# 3. Multiple Regression Analysis
# ----------------------------------------------------------

# Scatterplots and correlation matrix
reg_numeric_data <- pr_cleaned_data %>%
  select(
    log_review_time,
    pull_additions,
    pull_deletions,
    pull_changed_files,
    total_comments
  )

ggplot(reg_numeric_data, aes(pull_additions, log_review_time)) +
  geom_point(alpha = 0.4) +
  geom_smooth(method = "lm", se = FALSE)

ggplot(reg_numeric_data, aes(pull_deletions, log_review_time)) +
  geom_point(alpha = 0.4) +
  geom_smooth(method = "lm", se = FALSE)

ggplot(reg_numeric_data, aes(pull_changed_files, log_review_time)) +
```

```
  geom_point(alpha = 0.4) +
  geom_smooth(method = "lm", se = FALSE)

ggplot(reg_numeric_data, aes(total_comments, log_review_time)) +
  geom_point(alpha = 0.4) +
  geom_smooth(method = "lm", se = FALSE)

cor_matrix <- cor(reg_numeric_data, use = "complete.obs")
round(cor_matrix, 3)

# Regression model
reg_model <- lm(
  log_review_time ~ pull_additions +
    pull_deletions +
    pull_changed_files +
    total_comments,
  data = pr_cleaned_data
)

summary(reg_model)

# Diagnostic plots
par(mfrow = c(1, 3))
plot(reg_model, which = 1)
plot(reg_model, which = 2)
plot(reg_model, which = 3)
par(mfrow = c(1, 1))

# ---------------------------------------------------------
# 4. One-Way ANOVA
# ---------------------------------------------------------

# Boxplot comparison
ggplot(pr_cleaned_data,
       aes(issue_author_association, log_review_time)) +
  geom_boxplot()

# Group summary statistics
pr_cleaned_data %>%
  group_by(issue_author_association) %>%
  summarise(
    n = n(),
    mean = mean(log_review_time),
    sd = sd(log_review_time)
  )
```

```
# One-way ANOVA
anova_oneway <- aov(
  log_review_time ~ issue_author_association,
  data = pr_cleaned_data
)

summary(anova_oneway)

# Assumption checks
qqnorm(residuals(anova_oneway))
qqline(residuals(anova_oneway))

# ----------------------------------------------------------
# 5. Two-Way ANOVA
# ----------------------------------------------------------

anova_twoway <- aov(
  log_review_time ~ issue_author_association * pr_size,
  data = pr_cleaned_data
)

summary(anova_twoway)

# Interaction plot
interaction.plot(
  x.factor = pr_cleaned_data$pr_size,
  trace.factor = pr_cleaned_data$issue_author_association,
  response = pr_cleaned_data$log_review_time,
  fun = mean,
  type = "b",
  xlab = "PR Size",
  ylab = "Mean Log Review Time"
)

# Simple effects tests
t.test(
  log_review_time ~ pr_size,
  data = subset(pr_cleaned_data,
                issue_author_association == "Contributor")
)

t.test(
  log_review_time ~ pr_size,
  data = subset(pr_cleaned_data,
                issue_author_association == "Member")
)
```